Project 1: Applications of Fast Fourier Transform (*FFT)*

Instruction:

This is a group assignment (divide whole class into 6 groups).

Deadline: Feb 19, 2026

1) Purpose

Purpose: Apply the Fast Fourier Transform (FFT) to real data-science problems—**forecasting, classification, anomaly detection, denoising, and fast convolution**—while building a rigorous understanding of the **continuous** → **discrete** → **FFT** pipeline and its engineering trade-offs.

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2) Project Tracks (choose one of the suggested topics or explore one by yourself)

A. Time-Series Forecasting & Anomaly Detection

B. Audio / Vibration Classification (Predictive Maintenance or Speech Commands)

C. Image & 2D FFT (Denoising/Sharpening or Texture Features)

D. Compression & Signal Reduction for Big Data Streams
E. Biomedical Imaging: MRI Reconstruction Using FFT
F. FFT for Natural Disaster Monitoring (Earthquake, Landslide Signals)

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4) Deliverables

1. **Project Report (3–5 pages)**
2. **Presentation Slides** (7–15)
3. **Final Presentation (15–20 min)**: live demo + findings + limitations.